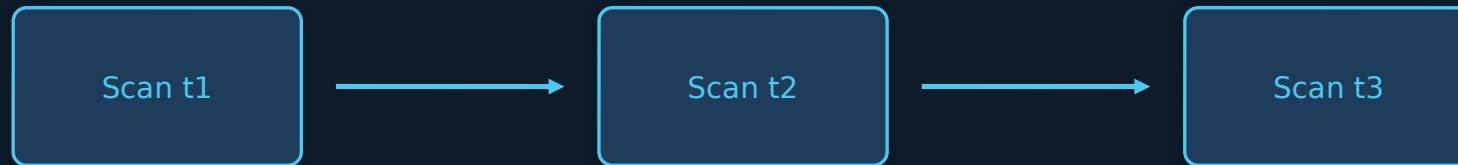


ADVISOR MEETING · MAY 22, 2026

Longitudinal & Generative Foundation Models for 3D Medical Imaging

A Research-Direction Survey & Proposal



one patient = an irregular sequence of 3D volumes

Songlin Zhao

Four Directions on the Table

1

TimeFM-3D

Longitudinal 3D imaging foundation model

MAIN LINE — DETAILED

2

Missing-data VLLM

Treating missing modality as a signal

SURVEYED

3

Diffusion-as-FM

A generative model reused as the backbone

SURVEYED

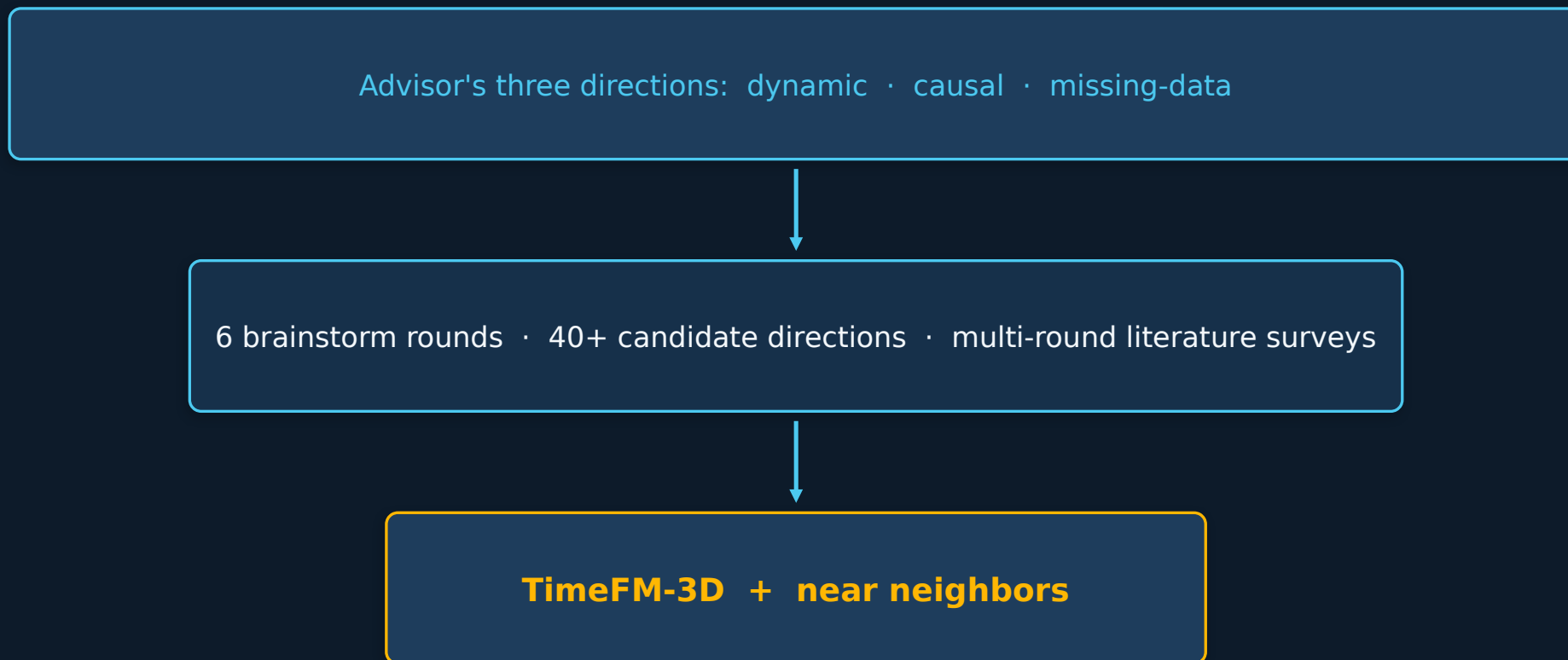
4

Causal VLLM

Counterfactual reasoning over imaging

BRIEF — NOT OUR STRENGTH

From Three Suggestions to One Converged Line



Honest read: the search has converged. Further divergence has near-zero marginal value — the bottleneck is now committing and running experiments, not finding more directions.

Patients Are Sequences, Not Snapshots

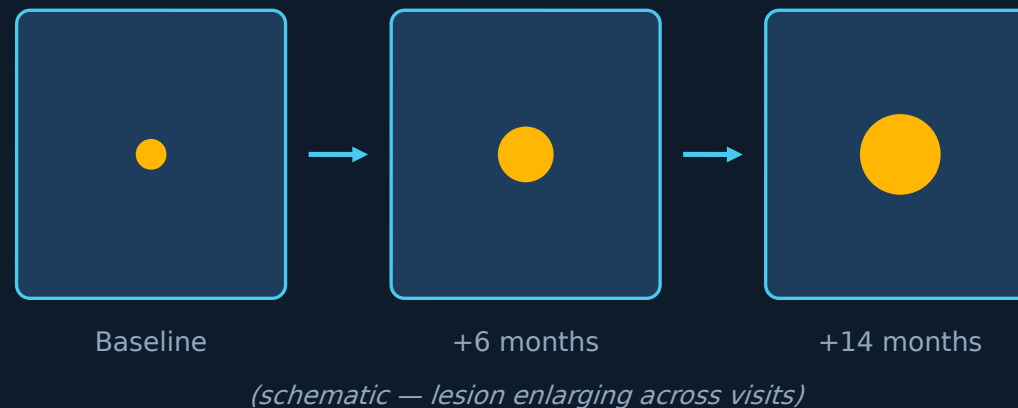
A follow-up is a **trajectory** of 3D CT/MRI over time.

Intervals are **irregular** — 3 mo, 7 mo, 14 mo.

The clinical question is always about **change**: progressing? responding? recurring?

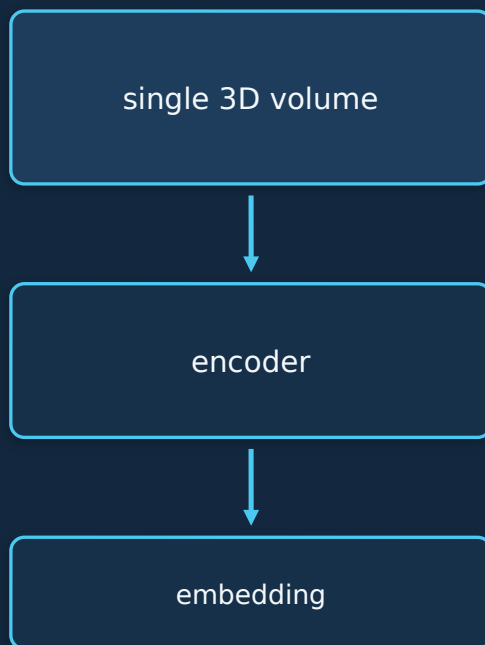
A single scan cannot answer it — the signal lives **between** timepoints.

Same patient — a brain metastasis followed over time



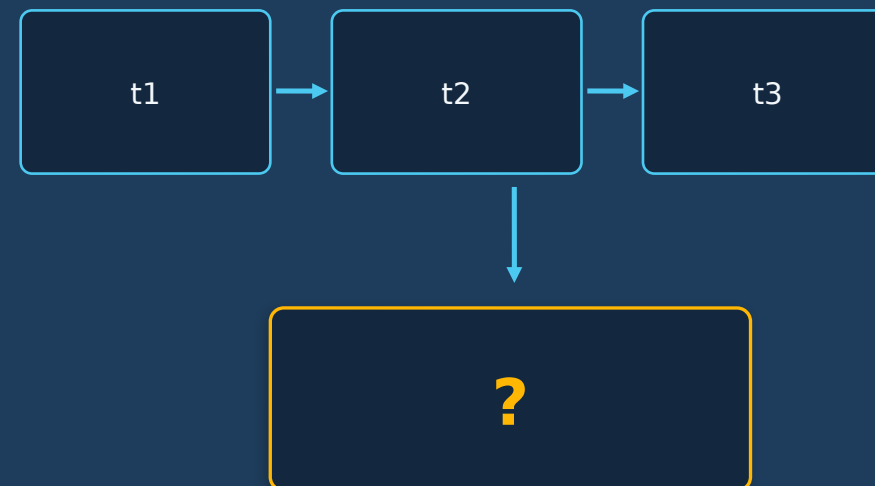
Current Foundation Models See One Timepoint

TODAY'S 3D MEDICAL FMs



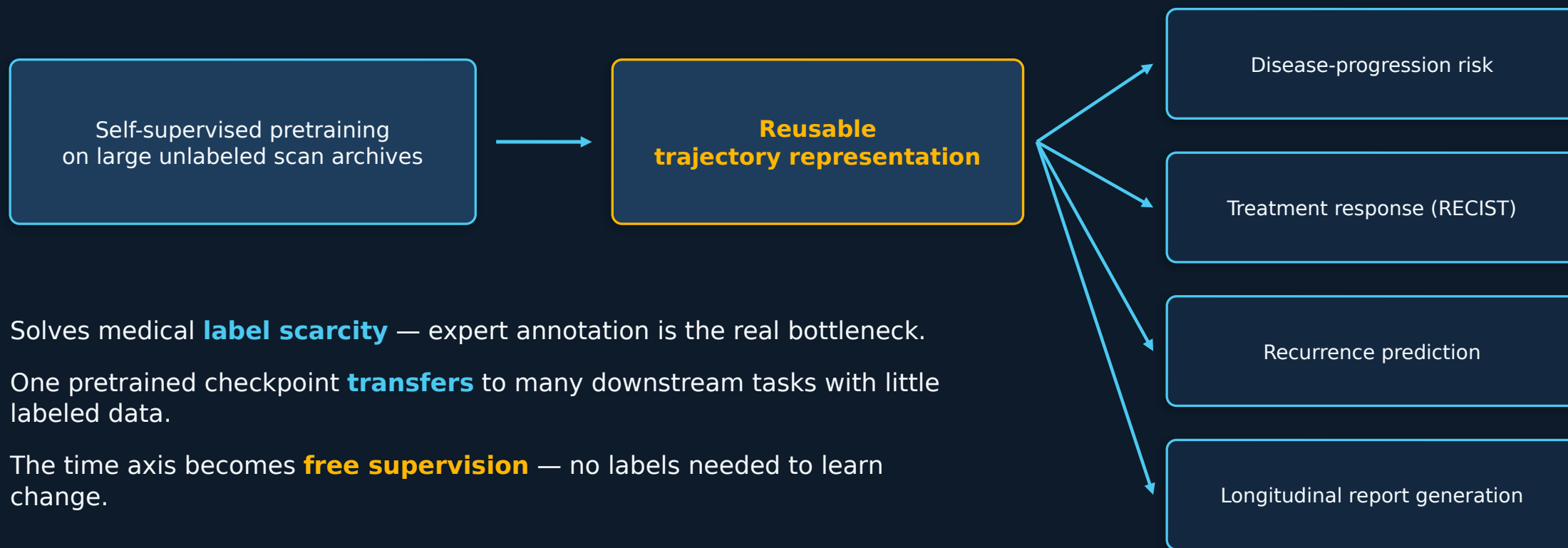
Merlin · CT-CLIP · CT-FM · RadFM · M3FM

WHAT IS MISSING



No FM natively ingests the sequence itself —
the time axis is the untaken white space.

Why a Foundation Model — Pretrain Once, Transfer Everywhere



Solves medical **label scarcity** — expert annotation is the real bottleneck.

One pretrained checkpoint **transfers** to many downstream tasks with little labeled data.

The time axis becomes **free supervision** — no labels needed to learn change.

one model → many clinical tasks

Survey — Who Else Is Working on This

#	Work	Input	Key limitation
1	TTE Pretraining (Stanford)	single CT	timeline lives in labels; image is not a sequence
2	SSL-AD	3D sequence	brain MRI only; ~3.2k patients; small
3	CRONOS	multi-timepoint	a forecaster, not an FM; no text / EHR
4	Temporal Flow Matching	3D sequence	generative, not a representation FM
5	Merlin (Nature 2026)	single CT + EHR	image static; timeline in EHR labels
6	BioViL-T / MAIRA-2 (Microsoft)	two chest X-rays	2D, not 3D — deep-dived next

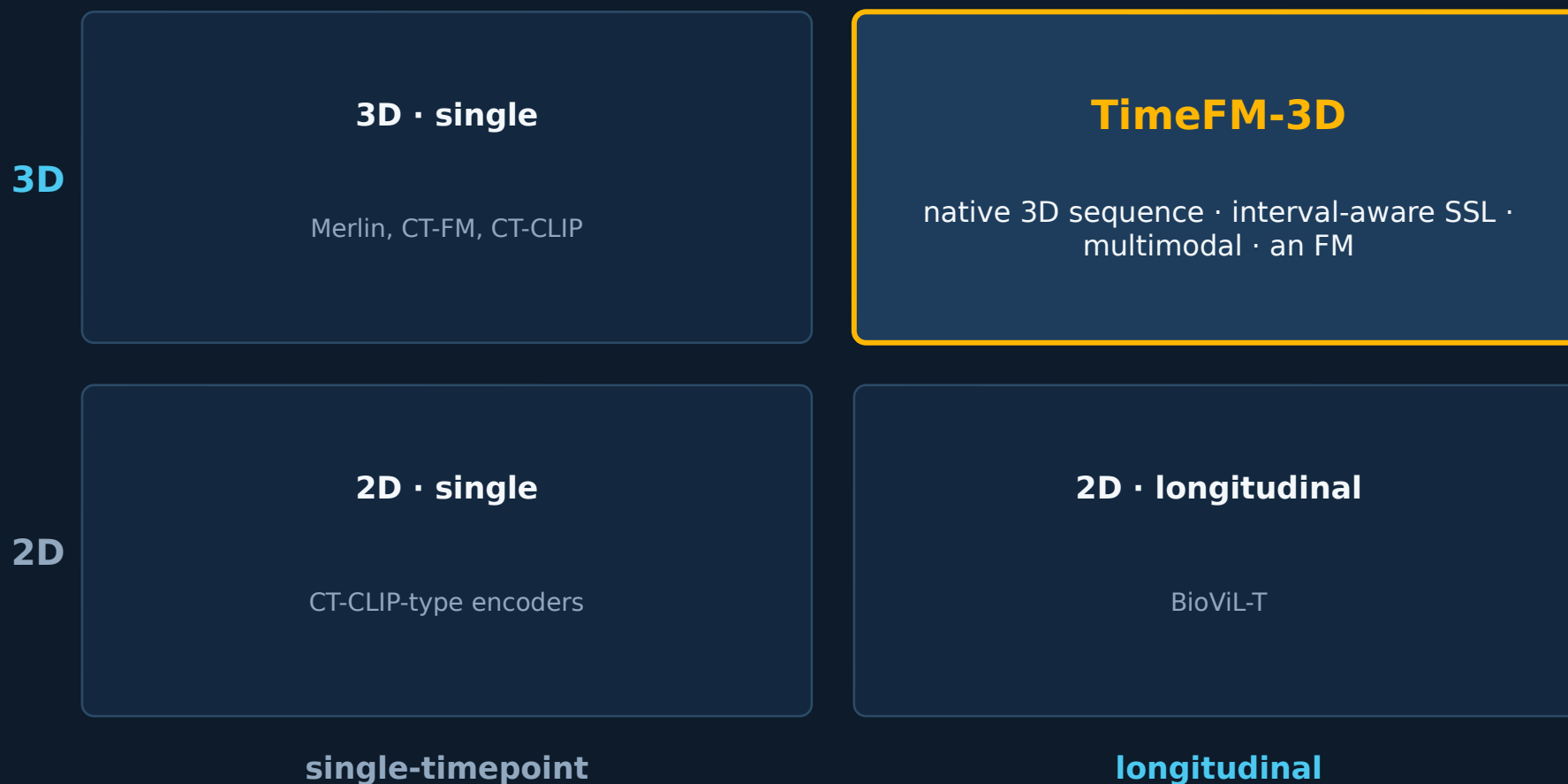
Nobody has shipped the full set: FM-scale + multi-organ + 3D sequence + interval-aware SSL + multimodal.

Where the Field Is Saturated — and Where It Is Empty

Sub-area	Maturity	Representative work
Single-timepoint 3D FM	Saturated	Merlin, CT-CLIP, CT-FM, RadFM, M3FM
Single image + EHR-label timeline	Taken (Stanford)	TTE Pretraining
2D longitudinal SSL	Mature (mostly CXR)	BioViL-T, MAIRA-2, L-MAE
3D longitudinal SSL	~ Empty — the gap	SSL-AD (brain-only, small)
3D longitudinal generation	Heating up	BrLP, SADM, Temporal Flow Matching

3D longitudinal self-supervised learning is essentially empty. That is the opening.

TimeFM-3D — The Untaken Quadrant



Deep-Dive — BioViL-T (Microsoft, CVPR 2023)

The closest 2D longitudinal vision-language FM.

(a) Prior VLP uses only a **single image** for InfoNCE contrastive learning.

(b) Discarding temporal links **weakens image-text alignment** (see the affinity matrix).

(c)(d) BioViL-T adds the **prior + current image** and does spatiotemporal modelling.

It learns temporal phrases — **“unchanged”**, **“worsening”** — and reaches SOTA.

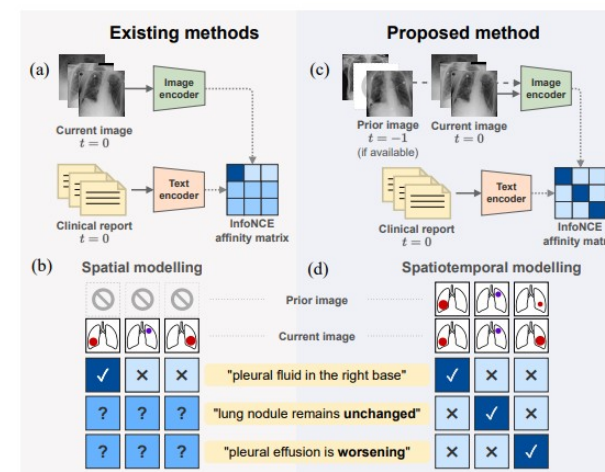


Figure 1. (a) Existing visual-language pre-training approaches [9, 32, 81] often use only a single image for contrastive learning (e.g., InfoNCE [49]). (b) In such settings, discarding the temporal connectivity of images limits the alignment of image-text pairs as shown with the affinity matrix, leading to suboptimal pre-training and missed opportunity to create additional model supervision for free. (c, d) Our approach exploits this domain knowledge by learning to incorporate a series of images and correlate them to reports, leading to pre-trained models that can generalise to a wider range of downstream tasks whilst achieving SOTA performance.

BioViL-T, Figure 1 (Microsoft, CVPR 2023)

BioViL-T Validates the Idea — in 2D

WHAT IT PROVES

Temporal links are **free extra supervision**.

Modelling change **improves representations** and downstream SOTA.

The longitudinal premise is **sound** — empirically demonstrated.

WHAT IT LEAVES OPEN

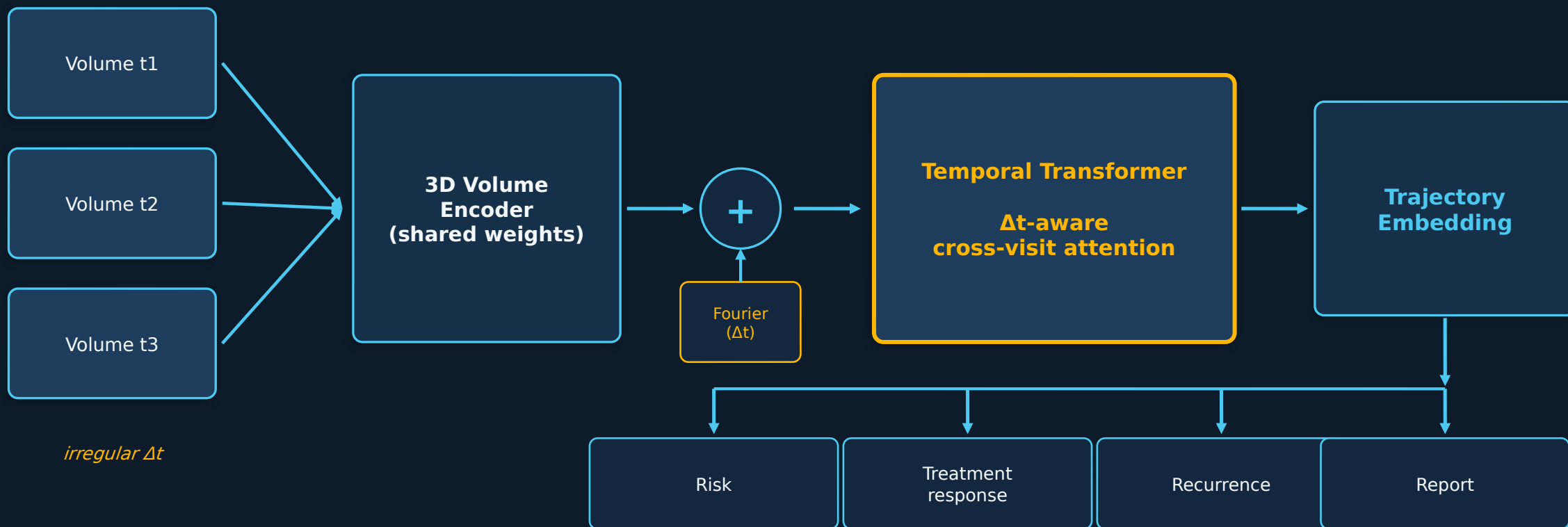
Only **2 timepoints** (prior + current).

Only **2D chest X-ray**, one anatomy.

No **3D volumes**, no irregular Δt , no multi-organ, no multimodal fusion.

TimeFM-3D = the same idea generalized to 3D · K timepoints · multi-organ · irregular intervals.

TimeFM-3D — Native Multi-Timepoint 3D Encoder

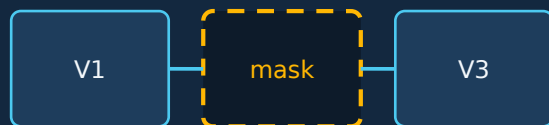


one trajectory embedding → many downstream clinical tasks

How It Learns — Three Self-Supervised Objectives

IA-MVM

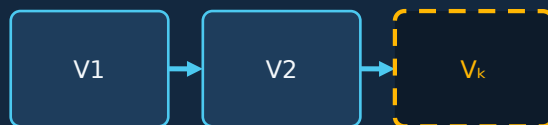
Interval-Aware Masked Volume Modeling



Mask a volume in the sequence, reconstruct it from neighbors. Short $\Delta t \rightarrow$ higher mask ratio; long $\Delta t \rightarrow$ predict the delta.

NVP-LS

Next-Volume Prediction in Latent Space



Given $V_1 \dots V_{k-1}$, predict the latent of the next volume V_k — forecasting as a pretraining signal.

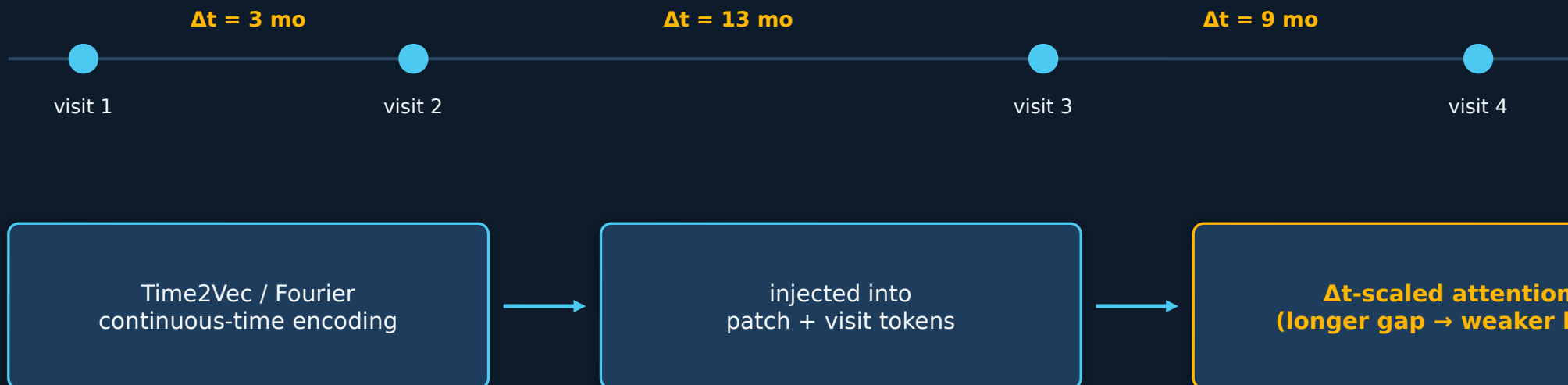
CMTC

Cross-Modal Temporal Contrast



Pair volume@t with its report / EHR window via InfoNCE; push other timepoints of the same patient apart.

Time Is Continuous and Irregular — Model It Explicitly



Naive sequence models assume **equal spacing** — wrong for real follow-up.

We encode the **absolute interval** so 3 months and 13 months are not treated alike. Borrowed from CRONOS / TaViT.

From Representation to Decision — Verifier-Guided Reasoning

PHASE 1 · SFT

clinical reports

extracted CoT trace

<thinking>
<answer>

PHASE 2 · RL (GRPO)

sample K
reasoning chains

verifier:
pathology · RECIST · mortality

reward →
update policy

Key: the verifier is a **real clinical outcome** — not a GPT-4 judge. This sidesteps the “reasoning-evaluation” quagmire and hits 2026 hiring keywords (reasoning · post-training · GRPO).

Data We Can Start With

Dataset	Patients	TP	Modality	Access
NLST	26,722 (LDCT arm)	3	chest LDCT	4-12 wk DUA
ADNI 1/2/3/4	~3,500+	2-10+	brain MRI + PET	days
OASIS-3	~1,378	multi	brain MRI	days
AIBL / PPMI	~350 / ~400	3+	brain MRI	days
Yale-Brain-Mets	1,430	mean 8	brain MRI	instant, no approval
CT-RATE	21,304 (~3-4k $\geq$ 2TP)	mostly 1	chest CT + reports	open

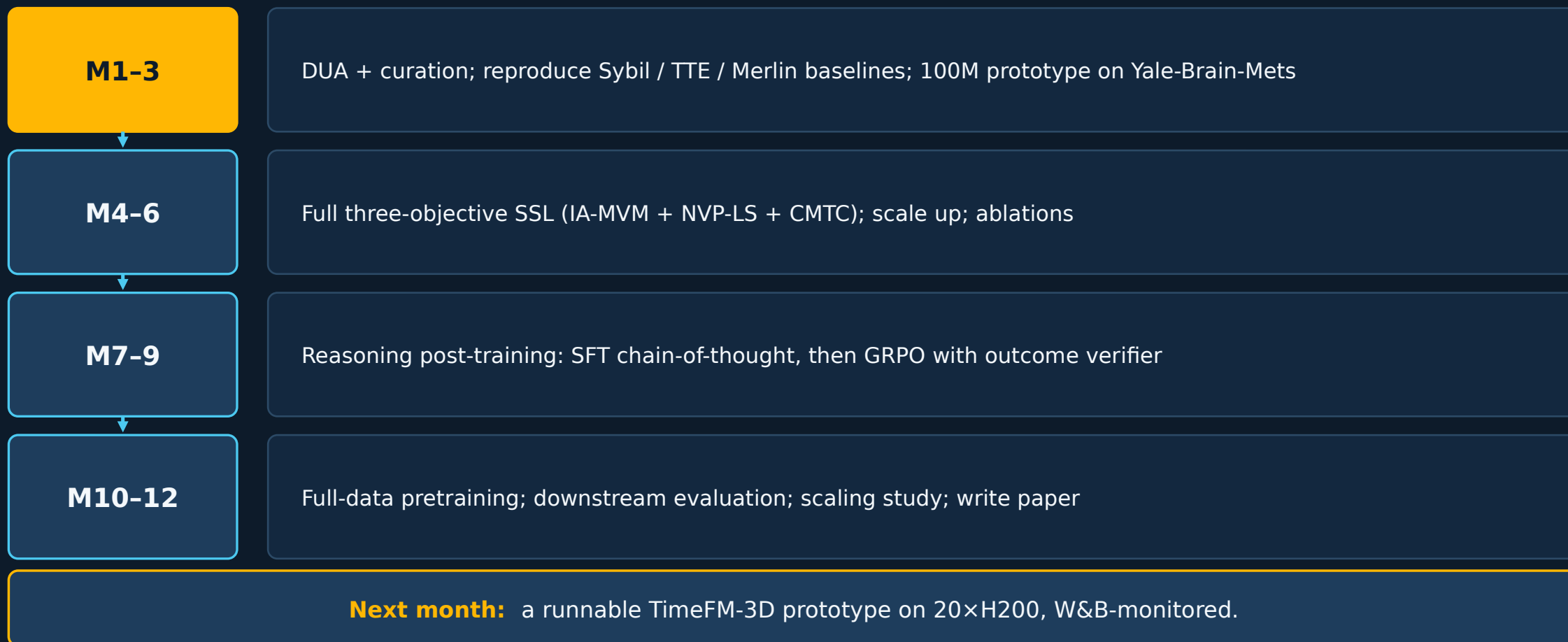
Prototype starts on Yale-Brain-Mets — instant download, zero approval, and it is brain imaging (hits the interest filter).

Data That Would Scale It Up — Pending Access

Dataset	Patients	TP	Note
Mayo Clinic (ours)	TBD	longitudinal	DUA in progress — the moat / unique multimodal cohort
UK Biobank repeat	~20k 2nd visit	2	paid DUA, 8–16 wk
NACC-SCAN	54k+	multi	multi-center brain imaging
ABCD	~11,800	≤4	adolescent brain, IRB-controlled
INSPECT (Stanford)	19,438	mostly 1	CTPA + 5-yr EHR — evaluation gold-standard
ISPY1/2 · LUMIERE · UPenn-GBM	small	4–7	treatment-response evaluation cohorts

Honest bottleneck: the constraint is data access and curation, not compute — 20×H200 is already in hand.

Preliminary Experiment & Timeline

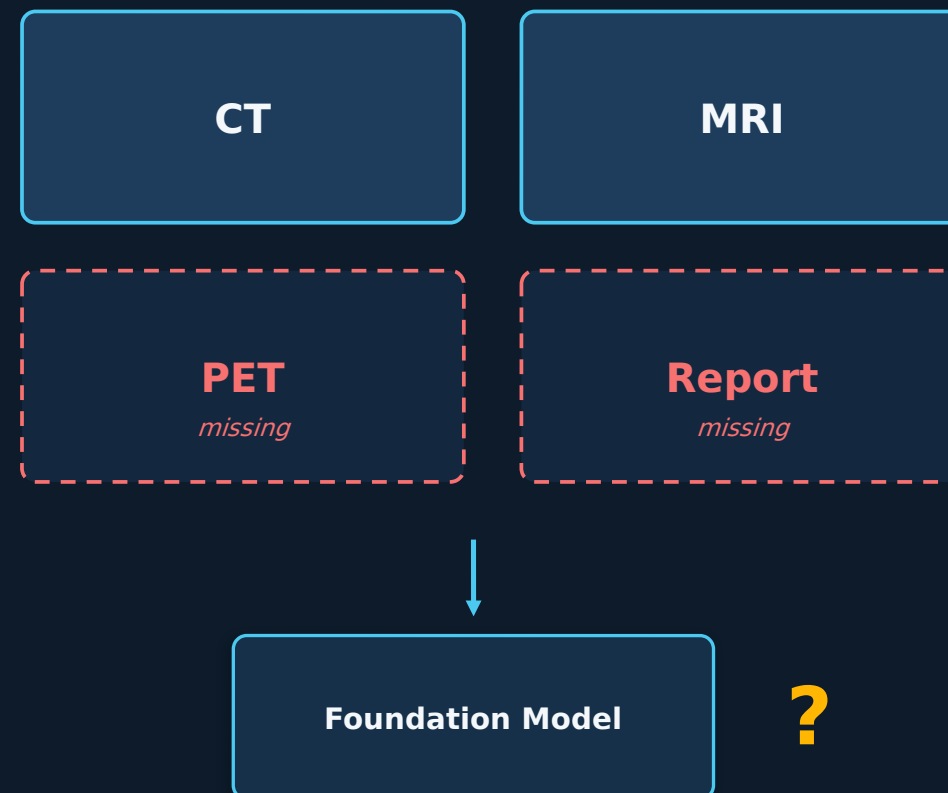


Direction 2 — When Modalities Are Missing

The advisor's **third direction**: a VLLM that handles missing data.

Real clinical data is **never complete** — no PET ordered, an MRI sequence skipped, a report absent.

The question: what does an FM **do** when a modality is missing?



Survey — What Is Solved, What Is Open

Sub-angle	Status
Method landscape (masked modeling, missing-aware prompts)	closed-set: largely solved
Uncertainty / calibration	taken — AECF (2025)
Cross-modal imputation	red ocean — PSNR-chasing
MNAR / informative missingness / causal view	OPEN — the gap
Benchmark / evaluation	MCAR assumption overstates robustness
Generative MLLM × missing data (medical)	half-empty

Six independent sub-surveys — five point to the same place: non-random missingness.

The Missing Pattern Itself Carries Information

MCAR VIEW — TODAY

modality missing



= random noise



drop it

MNAR VIEW — OPEN

modality missing



← clinical decision



a signal

No PET ordered → **the physician already had low suspicion of metastasis.** The absence is informative.

Filling In Missing Modalities Can Inject Bias

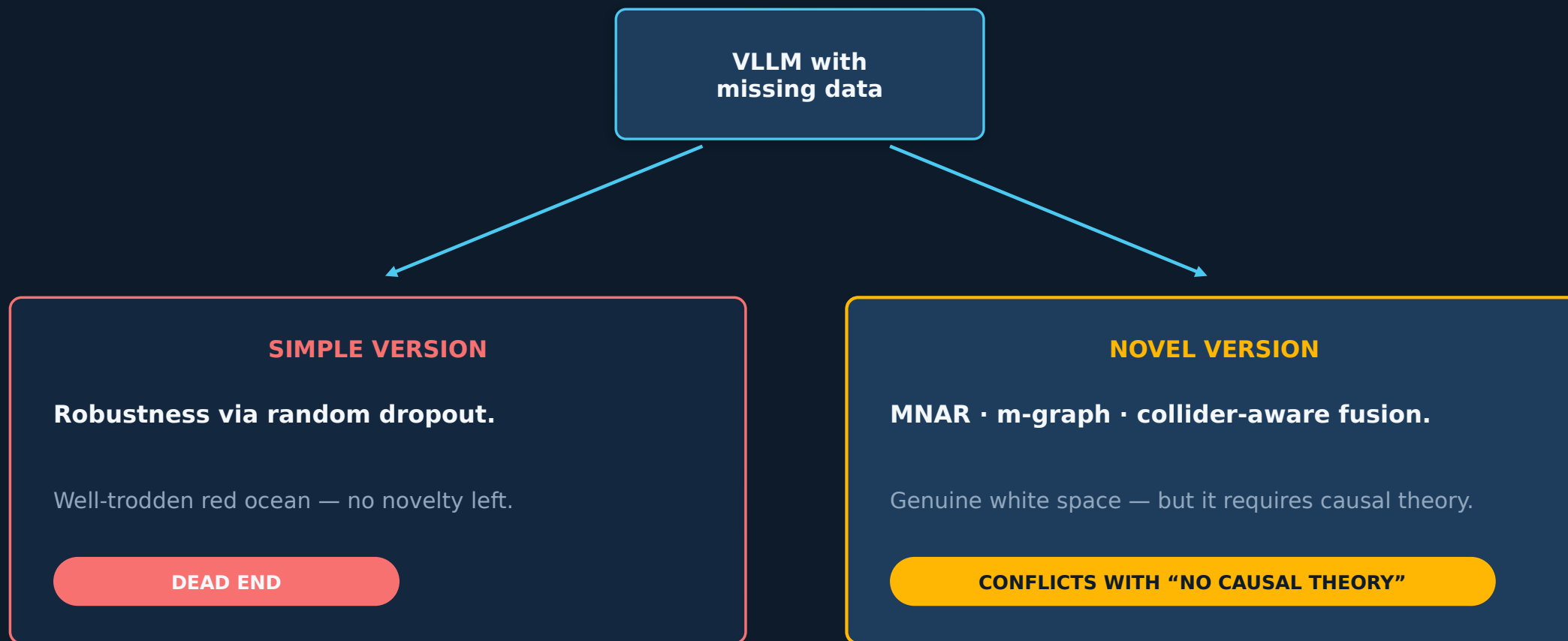


Under MNAR, naive imputation = **conditioning on a collider** → selection bias.

Hard-filling a tumor can **fabricate diagnostic evidence** that was never observed.

The fix is not better imputation — it is **modeling the missingness mechanism** itself.

The Honest Problem With This Direction



This tension is unresolved — it needs the advisor's input, not more brainstorming.

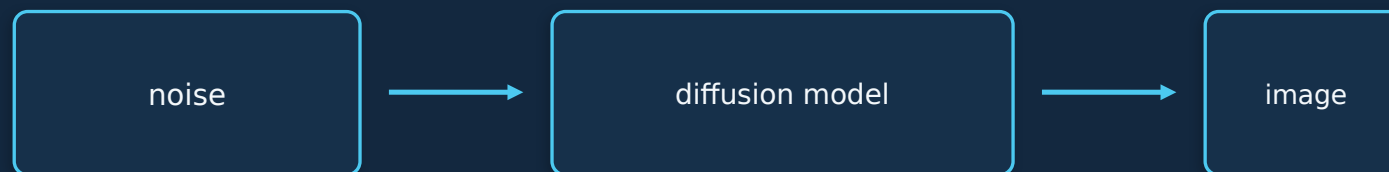
Where Direction 2 Stands

Option	Scope	Timeline	Risk
Fast	Missing-aware text prompt fed to an existing medical VLM	2–3 mo	low novelty
Benchmark	Realistic-missingness benchmark (ADNI / TCGA natural missingness)	3–4 mo	medium
High-ceiling	Full informative-missingness VLLM (needs causal)	8–12 mo	conflicts with “no causal”

Verdict: on hold — pending the advisor — which version of “missing data” did he mean?

Direction 3 — Can a Generative Model Be the Backbone?

DIFFUSION AS A GENERATOR



DIFFUSION AS A REPRESENTATION



Mainstream FMs use contrastive / MAE objectives. A different route: reuse a trained generative model's features for perception.

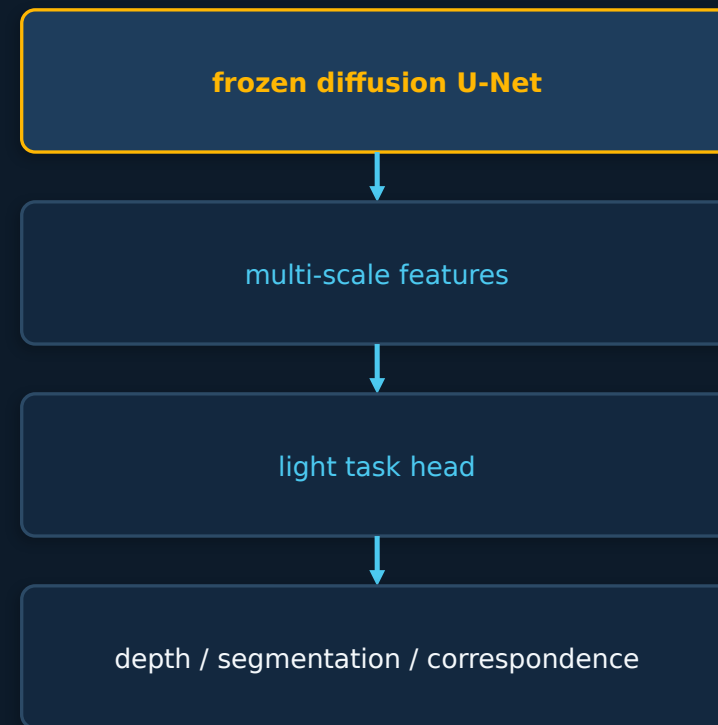
Marigold — Repurposing Diffusion Features

Marigold (CVPR 2024) took a diffusion model pretrained on natural images.

Frozen, plus a light head, it does **dense prediction** (e.g. depth) — beating specialized models.

The lesson: a **generative prior** is also a strong perception backbone.

The question: does this **transfer to medical 3D**?



no task-specific pretraining needed

Half-Empty, Not Empty

OCCUPIED

Li et al. (arXiv 2501.19265, Jan 2025)

frozen 3D medical diffusion → 3D CT organ segmentation.

Adjacent: DiffuGTS (CVPR 2025), LDAE.

STILL OPEN

Systematic head-to-head vs DINOv2 /
MAE, across classification + retrieval, on
CT and MRI.

The core paradigm — frozen 3D medical diffusion as a backbone — **is already taken**.

What is left is a **differentiating empirical study**, not a new paradigm.

Where Direction 3 Stands

Downgraded. Not a new paradigm — a MICCAI-scale empirical study differentiating from 2501.19265. Modest as a standalone flagship.

Diffusion-as-FM
probe

*evaluation
probe*

TimeFM-3D

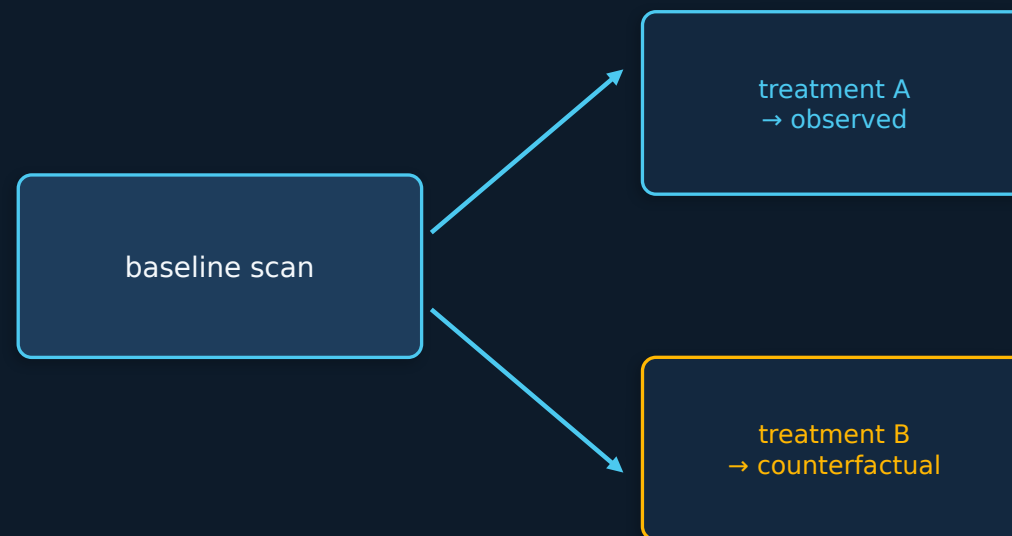
Best role: an evaluation probe /
benchmarking component for the
main line — not a separate project.

Direction 4 — Causal / Counterfactual VLLM

The advisor's **second direction**.

Asks a **counterfactual**: what would the imaging look like under a different treatment?

- Fits Mayo's longitudinal cohort well in principle.



observed vs never-observed outcome

Honest Assessment of Direction 4

Shelved. Causal inference has a high theory barrier — confounding, identifiability, semi-parametric efficiency. Not our current strength.

We are upfront: **we are not familiar with causal theory** and will not build on it as a foundation.

It could merge with Direction 2's m-graph — but that only **drags missing-data back into causal inference**.

Revisitable later; **not a starting point**.

Four Directions — Side by Side

Direction	Verdict	Why
TimeFM-3D	main line	real white space; fits brain + multimodal; can start now
Missing-data VLLM	on hold	the novel version needs causal theory — a conflict
Diffusion-as-FM	downgraded	core already occupied; good only as an evaluation probe
Causal VLLM	shelved	high theory barrier; not our current strength

There is no second direction on par with TimeFM-3D. That is the genuine bottom line of 40+ surveyed directions.

Three Things I Need to Confirm

1

What is actually in Liu's project?

Does “building on it” hold up — or is it effectively from scratch?

2

Which “missing data” did you mean?

Simple robustness has no novelty; the informative-missingness version needs causal theory.

3

Is “dynamic” exactly TimeFM-3D?

If yes — I start building immediately.

These answers are worth more than another round of brainstorming.

What I Will Do Next

1

Start Mayo DUA; download Yale-Brain-Mets and CT-RATE.

2

Run a TimeFM-3D prototype within two weeks (IA-MVM + NVP-LS + CMTC) on 20×H200.

3

Reproduce the Sybil / TTE / Merlin baselines.

4

Decide the missing-data scope once the advisor answers the three questions.

The direction has converged. Next is execution — and your answers to the three questions.